IoT solution: Smart Turtle Breeding System (STBS)

People: Farm Operators and Maintenance Technicians

Substitute:
What do people do without employing the solution?

- Farm operators have to manually measure the water quality and temperature in each breeding unit. (Farm Operators)
- Farm operators have to prepare feed portions and feed turtles according to daily schedules. (Farm Operators)
- Farm operators have to regularly inspect turtle activity, feeding behaviour, and health conditions. (Farm Operators)
- Farm operators have to inspect nesting areas and search for turtle eggs manually. (Farm Operators)
- Farm operators have to change water manually, while maintenance technicians have to schedule filter maintenance based on experience and inspection records. (Farm Operators and Maintenance Technicians)

Benefits:
What is good about the solution?

- The STBS aims to maintain suitable water conditions and reduce the duration of environmental abnormalities. (Farm Operators)
- The system can improve feeding consistency and reduce missed or inaccurate feeding. (Farm Operators)
- The system can help detect suspected abnormal activity or feeding behaviour earlier, allowing timely inspection. (Farm Operators)
- The system can identify suspected nesting activity and locations, reducing unnecessary searches for eggs. (Farm Operators)
- The system can reduce repetitive work and provide operating records for condition-based filter maintenance. (Farm Operators and Maintenance Technicians)

Features:
How does the solution work?

- The STBS continuously measures water temperature, pH, turbidity, and ammonia nitrogen, and automatically regulates water temperature. (Farm Operators)
- The system automatically dispenses scheduled feed portions and verifies delivery through sensor feedback. (Farm Operators)
- The system analyses video to identify turtle activity, feeding behaviour, and suspected nesting events, and stores recordings for verification. (Farm Operators)
- The system automatically operates pumps and valves for bounded partial water changes and verifies water level, flow, and post-action water quality. (Farm Operators and Maintenance Technicians)
- The system connects local controllers to the cloud through a gateway and provides remote monitoring, alerts, operation verification, and manual override. (Farm Operators and Maintenance Technicians)

Experience:
What does it feel like to use the solution?

- Farm operators will feel more confident because they can monitor all 20 breeding units from a central dashboard. (Farm Operators)
- Farm operators will feel less worried because alerts show abnormal readings and the corrective action taken by the system. (Farm Operators)
- Farm operators will feel more in control because they can verify automated operations and intervene when necessary. (Farm Operators)
- Farm operators will feel less burdened because they can review nesting videos and records before conducting targeted on-site checks. (Farm Operators)

Fears:
What are potential risks in employing the solution?
(E.g., what may not work, what may go wrong, etc.)

- There could be power or network failures that interrupt monitoring, automated control, or remote alerts. (Farm Operators and Maintenance Technicians)
- There could be sensor fouling, ageing, or calibration errors that lead to inaccurate readings and control decisions. (Farm Operators and Maintenance Technicians)
- The video analysis may falsely identify or miss abnormal behaviour and nesting events. (Farm Operators)
- There could be failures of feeders, heaters, pumps, or valves that disrupt feeding and breeding conditions. (Farm Operators and Maintenance Technicians)
- There might be high installation and ongoing costs for electricity, calibration, repairs, and consumables. (Farm Operators and Maintenance Technicians)

Needs:
What are the hidden needs and rational drivers for employing the solution?

- Farm operators need suitable water and temperature conditions for different turtle species and growth stages. (Farm Operators)
- Farm operators need to reduce repetitive inspection, feeding, and water-change work so that all 20 units can be managed efficiently. (Farm Operators)
- Farm operators need remote monitoring, timely alerts, and confirmation that automated corrective actions were effective. (Farm Operators)
- Farm operators need reliable activity, feeding, and nesting records to support on-site inspection and breeding management. (Farm Operators)
- Farm operators and maintenance technicians need an affordable, easy-to-maintain system with manual override and complete maintenance records. (Farm Operators and Maintenance Technicians)